

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403956
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Walentowski; Stephan

Steer-by-wire steering system

Abstract

A steer-by-wire steering system for a vehicle is disclosed, having a steering wheel, at least one sensor for capturing a steering angle of the steering wheel, a processing logic unit for determining a wheel setting angle requirement on the basis of the captured steering angle, and an actuator unit for setting a wheel setting angle at wheels of the vehicle according to the wheel setting angle requirement determined by the processing logic unit. The sensor is electrically connected directly to the vehicle electrical system and is supplied with electrical energy via the latter.

Inventors:	Walentowski; Stephan (Düsseldorf, DE)
Applicant:	ZF Automotive Germany GmbH (Alfdorf, DE)
Family ID:	88697033
Assignee:	ZF Automotive Germany GmbH (Alfdorf, DE)
Appl. No.:	18/323695
Filed:	May 25, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230382453 A1	Nov. 30, 2023

Publication Classification

Int. Cl.:	B62D15/02 (20060101); B62D5/04 (20060101)
U.S. Cl.:	
CPC	B62D15/0235 (20130101); B62D5/0463 (20130101);

Field of Classification Search

CPC: B62D (5/001); B62D (5/0235); B62D (5/0481)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6803736	12/2003	Hommel	318/806	G05B 9/02
2011/0118937	12/2010	Kariatsumari	701/41	B62D 5/0496
2020/0180685	12/2019	Suzuki	N/A	B62D 5/0412
2020/0307686	12/2019	Namikawa	N/A	B62D 5/0481
2020/0309566	12/2019	Fujita	N/A	G01L 5/0042
2021/0179166	12/2020	Bergmann	N/A	B62D 5/003
2022/0063710	12/2021	Tsubaki	N/A	B62D 5/0409
2022/0081023	12/2021	Miyake	N/A	B62D 5/0463
2022/0297747	12/2021	Polmans	N/A	B62D 5/0421

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10042308	12/2000	DE	N/A
102017217581	12/2018	DE	N/A
102019203111	12/2019	DE	N/A
102019007715	12/2020	DE	N/A

Primary Examiner: Tran; Long T

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to German Priority Application No. 102022205354.0, filed May 30, 2022, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The disclosure relates to a steer-by-wire steering system for a vehicle.

BACKGROUND

(3) Steering systems usually comprise a toothed rack which is mounted in a linearly displaceable manner in order to adapt a wheel position. Originally, such a toothed rack is coupled to the steering wheel via a steering rod, with the result that the toothed rack is linearly displaced by rotating the steering wheel.

(4) Motor vehicles having so-called steer-by-wire steering systems (SbW steering system), in which there is no longer a mechanical connection between the steering wheel and the toothed rack, will be increasingly used in future. The position of the steering wheel is captured electronically and the toothed rack is displaced accordingly by an electrical drive.

(5) On account of the omission of the mechanical connection between the steering wheel and the toothed rack, it is necessary for steer-by-wire steering systems to be particularly failsafe.

(6) Therefore, what is needed is a steer-by-wire steering system which is particularly failsafe.

SUMMARY

(7) According to the disclosure, by a steer-by-wire steering system for a vehicle, having a steering wheel, at least one sensor for capturing a steering angle of the steering wheel, a processing logic unit for determining a wheel setting angle requirement on the basis of the captured steering angle, and an actuator unit for setting a wheel setting angle at wheels of the vehicle according to the wheel setting angle requirement determined by processing logic unit, wherein the sensor is electrically connected directly to the vehicle electrical system and is supplied with electrical energy via the latter.

(8) The fact that the sensor is directly supplied with electrical energy via the vehicle electrical system means that the sensor is supplied with power independently of further components of the steering system, which contributes to a high degree of failure safety of the steering system.

(9) The processing logic unit comprises a function which makes it possible to determine the wheel steering angle requirement. In other words, the processing logic unit enables digital signal processing, of the signals received from the sensor.

(10) The processing logic unit may, in principle, be present in any desired control unit of the steering system as long as it is configured to receive signals from the sensor and to forward them to the actuator unit.

(11) In one exemplary arrangement, the at least one sensor is an angle sensor, for example, thus making it possible to directly determine an angular position of the steering wheel. However, it is also conceivable for the determination to be carried out in another manner in the case of other man-machine interfaces for capturing driver inputs, for example by a torque sensor or a force sensor.

(12) According to one exemplary aspect, the steering system comprises a steering unit, having a steering actuator for applying a torque to the steering wheel, wherein the steering unit is configured to determine a steering angle of the steering wheel on the basis of a position of the steering actuator. The steering angle is therefore captured redundantly, by the sensor on the one hand and by the steering unit on the other hand. This additionally increases the failure safety of the steering system. In one exemplary arrangement, since the steering unit and the sensor are supplied with electrical energy independently of one another, it is still ensured that the steering angle is captured if the sensor or the steering unit fails.

(13) The processing logic unit may be integrated in the steering unit and/or in the actuator unit. This means that the steering unit and/or the actuator unit has/have a corresponding control unit which is configured to capture and process the measured steering angle and to determine a corresponding wheel setting angle requirement.

(14) In one exemplary arrangement, both the steering unit and the actuator unit have an accordingly configured control unit, with the result that the processing logic unit is also redundantly present.

(15) However, only the actuator unit can cause the implementation of the wheel setting angle at the wheels. For this purpose, the actuator unit has, for example, a front axle actuator which can linearly displace a toothed rack.

(16) The steering unit and the actuator unit are connected to one another via a communication line, for example and the at least one sensor is connected directly to the actuator unit via a signal line. The actuator unit can therefore receive information relating to a steering angle in two mutually independent ways, from the sensor on the one hand and from the steering unit on the other hand. If the sensor or the steering unit fails, it is consequently ensured that a steering signal is still processed and the wheel setting angle can be set accordingly.

(17) The steering unit and the actuator unit are electrically connected directly to the vehicle electrical system and are supplied with electrical energy via the vehicle electrical system. This means that the at least one sensor, the steering unit and the actuator unit are completely independent of one another in terms of the supply with electrical energy. The supply with electrical energy is therefore particularly reliable.

(18) According to one aspect, the steering system has a first processing path and a second processing path, wherein each processing path comprises its own processing logic unit, an actuator

unit and at least one sensor for capturing the steering angle, and wherein the first processing path is a master path which, during normal operation, captures a steering angle and determines the wheel setting angle requirement ahead of the second processing path and sets the corresponding wheel setting angle. As a result of two processing paths being provided, each component is redundantly provided, thus additionally increasing the failure safety of the steering system. The processing logic unit in the first processing path is in particular identical to the processing logic unit in the second processing path.

(19) In one exemplary arrangement, more than two processing paths are provided, for example three processing paths. The failure safety is increased further by each additional processing path.

(20) The interconnection of the components in each processing path may be identical or different. In the case of a different interconnection, different failure scenarios may possibly be covered in both processing paths.

(21) The actuator units of the first and second processing path may each be connected to one another in terms of signaling via communication lines. The processing paths can therefore communicate with one another and interchange information if necessary.

(22) Each processing path may also have a steering unit, which steering units are likewise connected to one another in terms of signaling via communication lines.

(23) In one exemplary arrangement, the power supplies for the processing paths are independent of one another. To be precise, two vehicle electrical systems which differ from one another are provided, wherein each processing path is supplied with electrical energy by a different vehicle electrical system. This ensures that the steering system operates reliably even if the power supply for a processing path fails.

(24) The sensor of one of the two processing paths is, for example, electrically connected to the steering unit and is supplied with electrical energy via the latter. Consequently, the sensor is electrically connected directly to the vehicle electrical system in one of the two processing paths and to the steering unit in the other processing path. If the sensor is connected to the steering unit, it is possible to dispense with separate power conversion and power stabilization since this is already carried out in the steering unit. The complexity of the steering system can thus be reduced.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Further advantages and features of the disclosure will become apparent from the following description and from the accompanying drawings, to which reference is made. In the drawings:

(2) FIG. 1 schematically shows a steering system according to the disclosure,

(3) FIG. 2 schematically shows the electrical and electronic interconnection of the components of the steering system according to the disclosure,

(4) FIG. 3 schematically shows an electrical and electronic interconnection of a steering system according to the disclosure with two processing paths,

(5) FIG. 4 schematically shows an alternative electrical and electronic interconnection of a steering system according to the disclosure with two processing paths, and

(6) FIG. 5 schematically shows another alternative electrical and electronic interconnection of a steering system according to the disclosure with two processing paths.

DETAILED DESCRIPTION

(7) FIG. 1 schematically illustrates a steer-by-wire steering system **10** for a vehicle, having a steering wheel **12**.

(8) The steering system **10** comprises a toothed rack **14**. The toothed rack **14** is connected to a wheel carrier via track rods, which are not illustrated for the sake of simplicity, in such a manner that a linear displacement of the toothed rack **14** results in rotation of the wheel carriers about their

steering axis.

(9) The steering system **10** also comprises an actuator unit **20** for setting a wheel setting angle at the wheels **16**, **18** of the vehicle on the basis of a steering angle of the steering wheel **12**.

(10) A steering unit **22** is also provided and is configured to apply a torque to the steering wheel **12**. The steering unit **22** comprises a steering actuator **24** for this purpose.

(11) A sensor **26**, which in one exemplary arrangement, is an angle sensor, is configured to capture a steering angle of the steering wheel **12**.

(12) The steering system **10** also has a processing logic unit **28** which is configured to receive the steering angle captured by the sensor **26** and to determine a wheel setting angle requirement on the basis of the steering angle captured by the sensor **26**, on the basis of which wheel setting angle requirement a wheel setting angle is set by the actuator unit **20** at the wheels **16**, **18**.

(13) The processing logic unit **28** is configured to communicate with the sensor **26**, the actuator unit **20** and also the steering unit **22** using signaling.

(14) The processing logic unit **28** is schematically illustrated in FIG. **1**. However, the processing logic unit **28** is not a physical unit, but rather a signal processing function.

(15) The processing logic unit **28** may be integrated in the steering unit **22** or in the actuator unit **20** or a processing logic unit **28** may be integrated both in the steering unit **22** and in the actuator unit **20**, as explained in yet more detail below.

(16) In the exemplary arrangement illustrated, the actuator unit **20** comprises a servomotor **30** and a pinion **32** which acts on the toothed rack **14**.

(17) In an alternative exemplary arrangement, which is not illustrated for the sake of simplicity, the actuator unit **20** comprises electromotive control actuators instead of a toothed rack and a pinion **32**, wherein a control actuator is assigned to each wheel **16**, **18**.

(18) In the steer-by-wire steering system **10**, there is no mechanical coupling between the steering wheel **12** and the wheels **16**, **18**. Instead, a wheel setting angle is set by the servomotor **30**.

(19) For this purpose, the pinion **32**, which is in toothed engagement with the toothed rack **14**, is rotated by the servomotor **30**, as a result of which the toothed rack **14** is linearly moved.

(20) FIG. **2** illustrates an electrical and electronic interconnection of the components of the steering system illustrated in FIG. **1**, for example, the actuator unit **20**, the steering unit **22** and the sensor **26**.

(21) FIG. **2** also illustrates a wheel setting sensor **34** which is configured to measure a wheel setting angle of the wheels **16**, **18**.

(22) The actuator unit **20** can accordingly control the steering actuator **24** on the basis of a comparison of the wheel setting angle requirement with the wheel setting angle actually measured by the wheel setting sensor **34**.

(23) Both the actuator unit **20** and the steering unit **22** each have a control unit **36**, **38**.

(24) The control unit **36** of the actuator unit **20** is configured, for example, to control the servomotor **30** to set a desired wheel setting angle.

(25) In the exemplary arrangement according to FIG. **2**, the processing logic unit **28** is integrated in the actuator unit **20**, such as, for example, in the control unit **36**.

(26) In one exemplary arrangement, the control unit **36** is therefore configured to receive and process signals from the sensor **26** relating to the steering angle in order to determine a corresponding wheel setting angle requirement.

(27) For this purpose, the sensor **36** is connected to the actuator unit **20** via a signal line **40**.

(28) The control unit **38** of the steering unit **22** is configured to apply a torque to the steering wheel **12** by controlling the steering actuator **24** (see FIG. **1**).

(29) In addition, the steering unit **22**, more specifically, the control unit **38**, is configured to determine a steering angle of the steering wheel **12** on the basis of a position of the steering actuator **24**. For this purpose, the steering unit **22** may comprise a position sensor **39** (see FIG. **1**) which can transmit a corresponding signal to the control unit **38**.

- (30) If the sensor **26** fails, it is therefore still possible to determine a steering angle and to reliably and safely guide the vehicle.
- (31) In terms of capturing the steering angle, the steering unit **22** is thus used as a fallback solution for the sensor **26** or vice versa.
- (32) Position sensors are often designed such that they are reset to a value of zero when the vehicle is started. That is to say, if the vehicle was parked with the steering wheel turned, the position sensor **39** will not indicate the correct absolute steering angle, but rather only a change in the steering angle. In order to be able to nevertheless correctly determine an absolute steering angle by the position sensor **39**, a difference between the steering angle of zero measured by the position sensor **39** and the steering angle measured by the sensor **26** is calculated when the vehicle is started. This difference is stored until the next time the vehicle is parked and is added to the value measured by the position sensor **39**.
- (33) However, it is also conceivable for the steering angle to be able to be directly derived from the value measured by the position sensor **39**.
- (34) The steering unit **22** and the actuator unit **20** are connected to one another in terms of signaling via a communication line **42**, with the result that the steering angle determined by the control unit **38** can be transmitted to the actuator unit **20** or to the control unit **36** integrated in the actuator unit **20** and can be processed by the processing logic unit **28**.
- (35) It is also conceivable for a processing logic unit **28** to likewise be integrated in the steering unit **22**, more precisely in the control unit **38**. That is to say, the processing logic unit **28** is redundantly provided in the actuator unit **20** and in the steering unit **22**. In this case, a desired wheel setting angle can be transmitted directly to the actuator unit **20**.
- (36) As far as the supply with electrical energy is concerned, not only the actuator unit **20** and the steering unit **22** but also the sensor **26** are electrically connected directly to a vehicle electrical system **44**.
- (37) The vehicle electrical system **44** usually provides a voltage of 12 V.
- (38) Consequently, as far as the power supply is concerned, the stated components of the steering system **10** are independent of one another.
- (39) The steering system **10** according to FIG. 2 therefore still operates reliably if the sensor **26** fails or if the steering unit **22** partly or completely fails.
- (40) FIG. 3 illustrates an electrical and electronic interconnection of the components of the steering system **10** illustrated in FIG. 1 according to a further exemplary arrangement.
- (41) In this case, all components, for example, the actuator unit **20**, the steering unit **22**, the sensor **26** and the wheel setting sensor **34**, are provided in duplicate.
- (42) In one exemplary arrangement, the steering system **10** according to FIG. 3 has a first processing path **46** and a second processing path **48**, wherein each processing path comprises its own actuator unit **20**, a steering unit **22**, a processing logic unit **28** and at least one sensor **26** for capturing the steering angle.
- (43) The first processing path **46** is a master path which, during normal operation, captures a steering angle and determines the wheel setting angle ahead of the second processing path **48**.
- (44) Normal operation is used to mean operation in which all components of the steering system **10** are operating correctly.
- (45) The electrical and electronic interconnection in the master path corresponds to the interconnection illustrated in FIG. 2.
- (46) In the second processing path **48**, the interconnection of the sensor **26** differs, both in an electrical and in an electronic respect, from the interconnection of the sensor **26** in the first processing path **46**.
- (47) Specifically, the sensor **26** is electrically connected to the steering unit **22** and is therefore supplied with electrical energy via the steering unit **22**, rather than via the vehicle electrical system.
- (48) In terms of signaling as well, the sensor **26** in the second processing path **48** is connected to

the steering unit **22**. Information relating to the steering angle is consequently forwarded in the second processing path **48** to the actuator unit **20** via the steering unit **22**.

(49) The steering units **22** and the actuator units **20** of the two processing paths **46**, **48** may likewise communicate with one another via communication lines **50**.

(50) The steering system is particularly failsafe by virtue of two parallel processing paths **46**, **48** being provided.

(51) For example, the power supplies for the processing paths **46**, **48** are also independent of one another by virtue of the two processing paths **46**, **48** being supplied with electrical energy by different vehicle electrical systems **44**, **52**.

(52) For example, provision is made for the steering angle to be determined by the steering unit **22**, that is to say by capturing the position of the steering actuator **24** by the position sensor **39**, during normal operation.

(53) The wheel setting angle requirement is then determined by the processing logic unit **28**, that is to say either in the control unit **36** of the actuator unit **20** or in the control unit **38** of the steering unit **22**.

(54) The actuator units **20** then set a corresponding wheel setting angle of the wheels, which can be monitored by the wheel setting sensor **34**.

(55) A failure of the sensors **26** and of the steering unit **22** in the second processing path **48** has no consequence for the method of operation of the steering system **10**.

(56) If the position sensor **39** or the steering unit **22** as a whole in the first processing path **46** fails, there are various fallback options which ensure a reliable method of operation of the steering system **10**.

(57) On the one hand, the steering angle can be captured by the sensor **26** in the first processing path **46** and forwarded to the actuator unit **20**. The wheel setting angle requirement can be determined in the control unit **36** of the actuator unit **20** and the wheels can be correspondingly turned by the actuator units **20**. In an alternative exemplary arrangement, the capturing of the steering angle by the sensor **26** can be the variant implemented during normal operation. In this case, the position sensor **39** is correspondingly used as a fallback option.

(58) Alternatively, the steering angle can, in a similar manner, be determined in the steering unit **22** of the second processing path **48**, that is to say by capturing the position of the steering actuator **24**, instead of in the steering unit **22** of the first processing path **46**.

(59) Further alternatively, the steering angle can be captured by the sensor **26** in the second processing path **48** and can be forwarded to the steering unit **22** for further processing. This option is appropriate, in particular, when, for example, the position sensor **39** for capturing the position of the steering actuator **24** has failed, but the control unit **38** is still able to capture and at least forward the signal from the sensor **26**.

(60) In this case, the wheel setting angle requirement can be determined either in the control unit **38** of the steering unit **22** or, if the function of the latter is likewise impaired, in the control unit **36** of the actuator unit **20**.

(61) A particularly high degree of failure safety is achieved by the steering system **10** described in connection with FIG. **3**.

(62) For example, if the steering unit **22** in the first processing path **46** fails, at least two fallback options, which are completely independent of one another both in terms of the energy supply and in terms of the functional interconnection, remain. This means that a further partial failure does not result in the failure of all remaining fallback options.

(63) FIG. **4** illustrates a further electrical and electronic interconnection of the components of the steering system illustrated in FIG. **1**.

(64) The interconnection illustrated in FIG. **4** is virtually a mirror image of the interconnection illustrated in FIG. **3**.

(65) More precisely, the exemplary arrangement illustrated in FIG. **4** differs from the arrangement

illustrated in FIG. 3 in terms of the interconnection of the sensors 26.

(66) For example, the sensor 26 in the first processing path 46, which is the master path, is both electrically and electronically connected to the steering unit 22.

(67) The sensor 26 in the second processing path 48 is however electrically connected directly to the vehicle electrical system 52 and connected in terms of signaling to the actuator unit 20.

(68) If the steering unit 22 in the first processing path 46 fails in this case, the second processing path 48 is used as a fallback solution.

(69) If there is only a partial failure of the steering unit 22 in the first processing path 46, during which the steering unit 22 can still receive and forward signals from the sensor 26, the sensor 26 in the first processing path may also be used as a fallback option.

(70) Like in the exemplary arrangement described in FIG. 3, the sensor 26 in the first processing path 46 can also be used here to capture the steering angle during normal operation.

(71) FIG. 5 illustrates a further electrical and electronic interconnection of the components of the steering system illustrated in FIG. 1.

(72) In the exemplary arrangement illustrated in FIG. 5, the electrical and electronic interconnection is identical in both processing paths 46, 48.

(73) More precisely, the sensors 26 of both processing paths 46, 48 are directly connected to the respective vehicle electrical system 44, 52.

(74) Both sensors 26 are connected, via signal lines 40, to the respective actuator unit 20 in the corresponding processing path 46, 48.

(75) The exemplary arrangement illustrated in FIG. 5 has a particularly high degree of failure safety since the sensors 26 in both processing paths 46, 48 are electrically independent.

Claims

1. A steer-by-wire steering system for a vehicle, comprising a steering wheel, at least one sensor for capturing a steering angle of the steering wheel, a processing logic unit for determining a wheel setting angle requirement on the basis of the captured steering angle, an actuator unit for setting a wheel setting angle at wheels of the vehicle according to the wheel setting angle requirement determined by the processing logic unit, a steering unit comprising a steering actuator for applying a torque to the steering wheel, wherein the steering unit is configured to determine a steering angle of the steering wheel on the basis of a position of the steering actuator, wherein the steering unit provides redundant steering angle detection independent from the at least one sensor, wherein the at least one sensor is electrically connected directly to an vehicle electrical system and is supplied with electrical energy via the vehicle electrical system, wherein the processing logic unit is configured to determine a difference between a steering angle of zero measured by a position sensor and the steering angle measured by the at least one sensor when the vehicle is started, store the difference, and add the difference to subsequent values measured by the position sensor to determine an absolute steering angle.

2. The steer-by-wire steering system as claimed in claim 1, wherein the at least one sensor is an angle sensor.

3. The steer-by-wire steering system as claimed in claim 1, wherein the processing logic unit is integrated in the steering unit and/or in the actuator unit.

4. The steer-by-wire steering system as claimed in claim 1, wherein the steering unit and the actuator unit are connected to one another via a communication line and the at least one sensor is connected directly to the actuator unit via a signal line.

5. The steer-by-wire steering system as claimed in claim 1, wherein the steering unit and the actuator unit are electrically connected directly to the vehicle electrical system and are supplied with electrical energy via the vehicle electrical system.

6. The steer-by-wire steering system as claimed in claim 1, wherein the steering system has a first

processing path and a second processing path, wherein each processing path comprises its own processing logic unit, an actuator unit and at least one sensor for capturing the steering angle, and wherein the first processing path is a master path which, during normal operation, captures a steering angle and determines the wheel setting angle requirement ahead of the second processing path.

7. The steer-by-wire steering system as claimed in claim 6, wherein the actuator units of the first and the second processing path are each connected to one another in terms of signaling via communication lines.

8. The steer-by-wire steering system as claimed in claim 6, wherein the power supplies for the processing paths are independent of one another.

9. The steer-by-wire steering system as claimed in claim 6, wherein the sensor of one of the two processing paths is electrically connected to a steering unit and is supplied with electrical energy via the steering unit.
